

**LAPORAN PRAKTIKUM
WORKSHOP SISTEM INFORMASI WEB SERVER
(TIF330805)
SEMESTER III**



**Acara 13
Service Layer, Validasi, Session, dan Business Logic**

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```

class MahasiswaService
{
    private MahasiswaRepository $repo;
    private ProdiRepository $prodiRepo;

    public function __construct(MahasiswaRepository $repo,
ProdiRepository $prodiRepo)
    {
        $this->repo      = $repo;
        $this->prodiRepo = $prodiRepo;
    }
}

```

1. Membuat MahasiswaService dan Inject MahasiswaRepository dan ProdiRepository melalui constructor.

```

// ----- Validasi (private) -----
private function normalize(array $in): array
{
    return [
        'nim'      => trim($in['nim'] ?? ''),
        'nama'     => trim($in['nama'] ?? ''),
        'email'    => trim($in['email'] ?? ''),
        'prodi_id' => (int) ($in['prodi_id'] ?? 0),
        'angkatan' => (int) ($in['angkatan'] ?? date('Y')),
        'status'   => $in['status'] ?? 'aktif',
    ];
}

```

2. Proses validasi data mahasiswa di MahasiswaService.

```

// ----- Proses tambah -----
public function create(array $input): array
{
    $input = $this->normalize($input);
    [$mhs, $errors] = $this->fill(new Mahasiswa(), $input);

    if ($errors) {
        return $this->fail($input, $errors);
    }

    if ($this->repo->find($mhs->getNim()) !== null) {
        return $this->fail($input, [], 'NIM sudah
terdaftar. ');
    }

    try {
        $this->repo->create($mhs);
    }
}

```

```

        } catch (PDOException $e) {
            return $this->fail($input, [], 'Data gagal
disimpan.');
```

3. Method create ()

```

// ----- Proses ubah -----
public function update(array $input): array
{
    $input = $this->normalize($input);
    $mhs   = $this->repo->find($input['nim']);

    if ($mhs === null) {
        $result = $this->fail($input, [], 'Data mahasiswa
tidak ditemukan.');
```

// NIM

adalah kunci, tidak diubah

4. Method Update ()

```

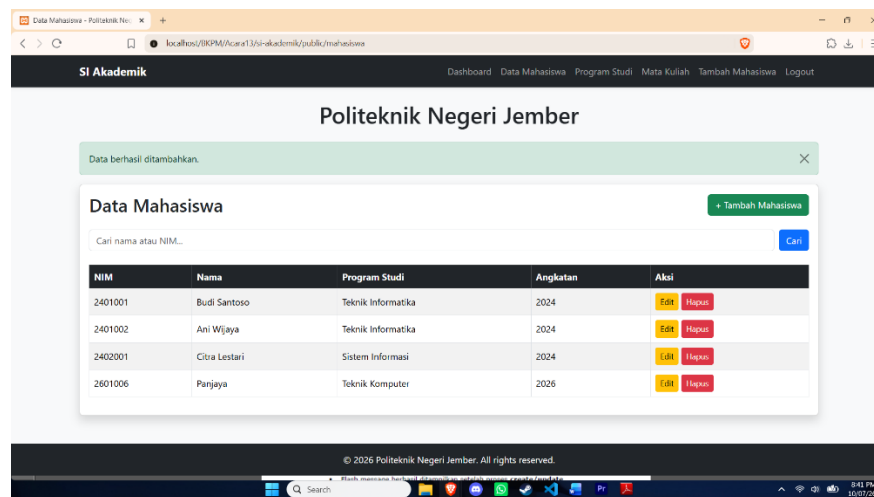
<?php $flash = Flash::pull(); // hanya tampil sekali, Langsung
dihapus dari session ?>
<?php if (!empty($flash)): ?>
```

```

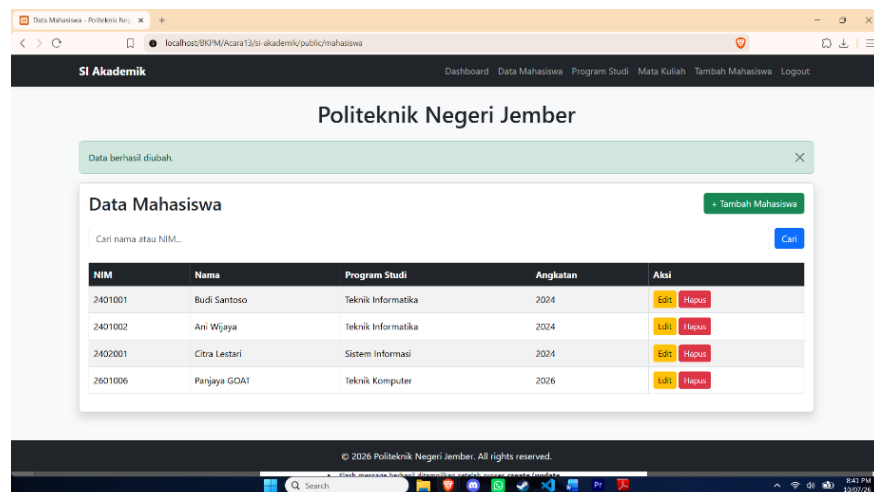
<div class="alert alert-<?php echo htmlspecialchars($flash['type'])
?> alert-dismissible fade show" role="alert">
    <?php echo htmlspecialchars($flash['message']) ?>
    <button type="button" class="btn-close" data-bs-
dismiss="alert" aria-label="Close"></button>
</div>
<?php endif; ?>

```

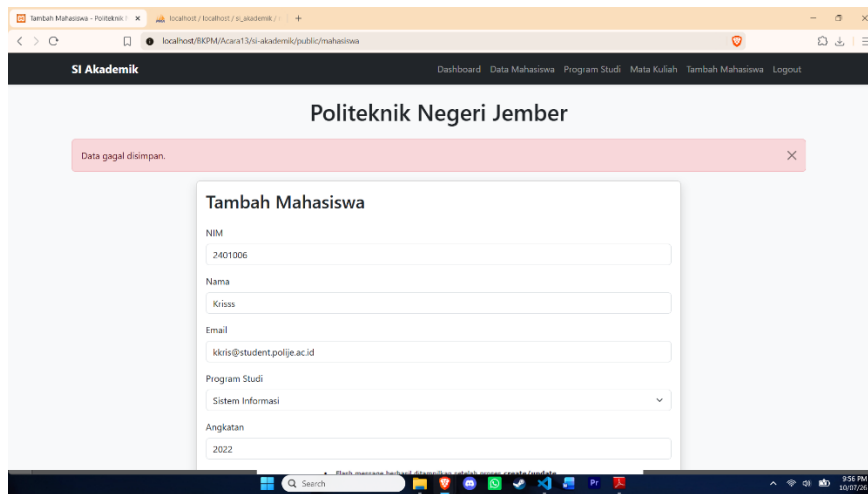
5. Flash Message menggunakan Session



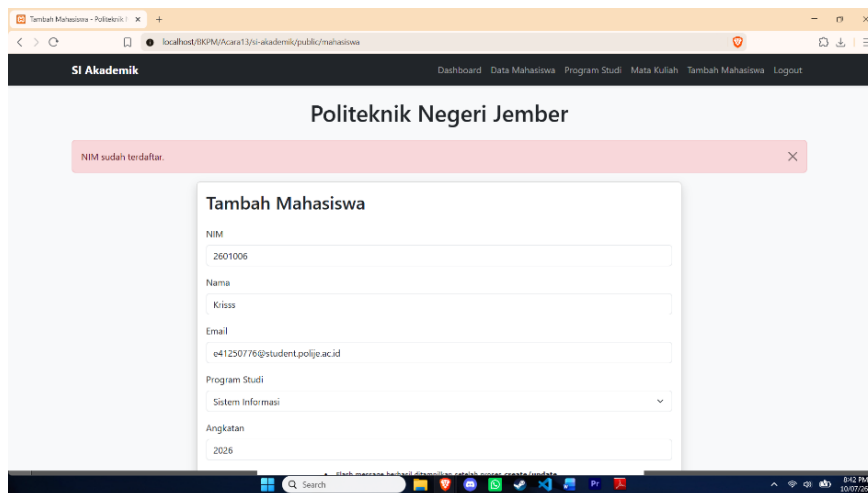
6. Tampilan Pesan Data berhasil ditambah



7. Tampilan Pesan Data berhasil Diubah



8. Tampilan Pesan Data gagal Disimpan



9. Tampilan Pesan NIM Sudah terdaftar.